Ajay Sharma Mandadi

amandadi3@gatech.edu | https://www.linkedin.com/in/ajay-mandadi

Introduction

With a strong research background in computer architecture, high-performance computing, and hardware-software co-design, I build, tinker and analyze novel computing stacks. My research spans from energy-efficient analog systems for high-precision, error-tolerant computation to near-memory architectures that accelerate graph workloads.

Education

Doctor of Philosophy, Electrical and Computer Engineering Jan 2023 – Present
Georgia Institute of Technology
Energy Efficient Architectures for High Performance Computing
Advisor: Dr. Tom Conte.

Master of Science in Electrical and Computer Engineering Aug 2021 - Dec 2022
Georgia Institute of Technology
Computer Systems and Software

Bachelor of Technology in Electronics and Communication Engineering Aug 2014 – May 2018
R.V. College of Engineering

Work Experience

GPU Subsystems Architecture Intern, AMD – Orlando, FL Aug 2022 – Dec 2022
Project: Evaluating performance bottlenecks in the GPU front-end (command processor)
Supervisor: Rex McCrary

- Evaluated a wide range of workloads using a RISC-V based GPU simulator, conducting detailed trace analysis and architectural profiling to classify bottlenecks.
- Worked on micro-architectural improvements and design optimizations while collaborating closely with the core architecture team within the Radeon Technology Group (RTG).

Operating System Software Engineer Intern, Intel – Folsom, CA May 2022 – Aug 2022
Project: Enhancing IOMMU diagnostic toolkit
Supervisor: Dr. Farhan Habib

- Improved debugging infrastructure and enhanced hardware/software interfacing of the Input/Output Memory Management Unit (I/O MMU) as part of the Microsoft Technology Centre (MTC) team.
- Earned recognition for exceeding expectations at the internship project showcase.

Software Engineer II, Cisco – Bangalore, India Sept 2018 – Jul 2021
Supervisor: Rekha Puvvada

- Designed key features for Next Generation Firewalls (NGFW) in the Cisco's Security Business group.
- Lead developer in a team of 3 - Packet tracer debugability tool for Cisco's NGFW.
- Pioneered various important features end-to-end as part of the Data Acquisition(DAQ) layer of Snort.
- Contributed immensely to building an SSL+TCP proxy stack and implemented the newer TLS 1.3 standard.

Research Experience

Graduate Research Assistant, Georgia Institute of Technology Jan 2023 – Present

1. **Using Residue Number System (RNS) for Analog Iterative Solvers:**

   Designing a framework for representing high-precision floating-point values in compact, analog crossbar-friendly formats and error handling strategies for energy efficiency and performance.

2. **Hardware-Software Co-design for Graph Acceleration (FORZA):**

   Developed a communication-centric, data focused architecture enabling massive fine-grained parallelism for accelerating irregular graph workloads.

3. **Fully Homomorphic Encryption (FHE) and its Architectural Challenges:**

   Evaluated CKKS variant of FHE to understand/survey its compute and memory implications.

Teaching Experience

Head Teaching Assistant, High Performance Computer Architecture Jan 2025 – May 2025
CS 6290/4290, ECE 6100/4100, Georgia Institute of Technology

- Supported over 150 graduate and undergraduate students in lectures, labs, and office hours.
- Authored all homework, exams and projects, ensuring alignment with course objectives/learning outcomes.

Research Mentor, TINKER Lab Aug 2025 – Present
Georgia Institute of Technology

- Supervise undergraduate/graduate students on research directions, algorithms and design.

Publications & Posters

1. **ARIT: An Analog RNS-based Iterative Solver for Extending Native Crossbar Precision**
   A. S. Mandadi, P. Gupta, T.M. Conte.
   33rd IEEE International Symposium on High-Performance Computer Architecture (HPCA 2027).
   (To appear in March 2027)

2. **Distributed SpMV with HClib-Actor: A Case Study in Alternatives to MPI-Based PETSc.**
   D. W. Pang, A. S. Mandadi, S. P. Singhal, P. Gupta, V. Sarkar, A. Hayashi, T.M. Conte.
   SC Workshops '26: Proceedings of the SC '26 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis. (To appear in November 2026)

3. **Breaking All-Reduce Bottlenecks with Asynchronous Influence Maximization at Scale.**
   Shubhendra Pal Singhal, Ajay Sharma Mandadi, Akihiro Hayashi, Thomas M Conte, Vivek Sarkar.
   2026 IEEE 26th International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)
   doi: 10.1109/CCGridW69005.2026.00025 (Publication).

4. **Optimizing Sparse/Dense VEGETA Accelerator Performance with Microscaling Quantization.**
   K. B. Nine, C. Talley, A. S. Mandadi, T. Krishna and A. Raychowdhury.
   IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025.
   doi: 10.1109/ISCAS56072.2025.11043957 (Publication).

5. **The FORZA Architecture Execution Model**
   A. S. Mandadi, P. Gupta, S.Cheng, A. Hayashi, J. Young, T. Conte, and V. Sarkar.
   Center for Research into Novel Compute Hierarchies (CRNCH) Summit, Atlanta, GA, 2024 (Poster).

6. **An Introduction to the FORZA Architecture**
   S. Cheng, P. Gupta, A. S. Mandadi, J. Young, T. Conte, and V. Sarkar.
   Center for Research into Novel Compute Hierarchies (CRNCH) Summit, Atlanta, GA, 2024 (Poster).

Invited Talks

"***Homomorphic Encryption and its Computing Challenges***"
Guest Lecture, Georgia Tech, CS/ECE 6290/6100 HPCA/ACA (Host: Prof. Tom Conte)

Awards

- **Ph.D. Fellowship**, Center for Research into Novel Compute Hierarchies 2025-26
- NSA Codebreaker Challenge **High Performer**, NSA 2021
- **Champion Early-in-Career Network Director Award**, Cisco Systems 2020